Alexander Bernstein

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Education

University of California, Santa Barbara Santa Barbara, CA
Ph.D. in Statistics *Sept. 2025*
Dissertation: "Long-Only Minimum Variance Portfolio Composition for Factor Models"

Washington University in St. Louis St. Louis, MO
M.Sc. in Systems Science and Applied Mathematics *Dec. 2016*
B.A. in Mathematics and Economics (Cum Laude) *May 2014*

Research Interests

Convex and portfolio optimization, factor models and feature construction, asset pricing (options and bonds), multivariate statistics, time-series analysis, probability and stochastic analysis, and applications of machine learning in financial mathematics.

Publications and Working Papers

Banerjee, T., **Bernstein, A.**, Feinstein, Z., (2025). "Dynamic clearing and contagion in financial networks". *European Journal of Operational Research* 321.2, pp. 664–675.

Bernstein, A., Shkolnik, A., (2025). "Asymptotics of Quadratic Forms on a Simplex". In Preparation.

Talks and Presentations

Analytical Solutions to the Constrained Markowitz Problem via Fixed Point Theory
INFORMS Annual Meeting, Phoenix, AZ *Oct. 2023*

Explicit Solutions for Position Constrained Minimum Variance Portfolios
SIAM Conf. on Applied and Computational Discrete Algorithms, Online Poster *July 2021*
CDAR Risk Seminar, UC Berkeley, Berkeley, CA *March 2020*

Projects

Mean Field Game Reinforcement Learning: Implemented a **Q-Learning** algorithm to solve the major-minor cyber-security game (Carmona, Delarue, et al. 2018)[Ex. 7.2.3]. Learned optimal defense policies through interaction with a dynamic population distribution μ.

Conditional Covariance Estimation: Built a **Covariance Regression** model (**CovRegpy**) for covariate-adjusted covariance of idiosyncratic equity returns, a parsimonious alternative to **MGARCH**. Leveraged exogenous covariates to improve stability and reduce parameter overhead in high-dimensional settings.

Quantitative Options Pricing: Implemented **Monte Carlo** (Euler/Milstein) and **finite-difference** (implicit, explicit, and Crank-Nicolson) solvers for the **Heston** and **CEV** models. Built **Gaussian Process** (Matérn) and **PyTorch** neural-network surrogates to recover prices and Greeks (via autograd) at lower cost.

Factor-Model Portfolio Pipeline: Architected a modular, spec-driven (**YAML**) **Python** pipeline for portfolio construction from a universe of S&P 500 assets. Builds **PCA** factor models with pluggable covariance estimators (**POET** thresholding, eigenvector shrinkage, GARCH) and return estimators (**Numba**-JIT **Kalman**/EKF filters, robust Student-t, Black-Litterman, L1-adaptive control) registered through decorator registries. Solves constrained mean-variance optimization via **CVXPY** (CLARABEL) with a walk-forward out-of-sample backtester and a CLI rebalancer that emits live trades. Built on clean ABCs with unit tests and reproducible configs; runs the full **S&P 500** (daily/weekly data since 2010) in ∼5–10 min per strategy.

Portfolio Aggregator: Built and self-hosted a **Dockerized FastAPI** dashboard that aggregates multi-brokerage accounts through a third-party **REST API** (SnapTrade) into a SQLite store, with scheduled auto-sync, historical snapshots, and daily-change tracking.

Wellbore Geology Prediction (Kaggle): Built the geologic-surface stage of a two-stage pipeline predicting subsurface layer depth along horizontal wells: **SPDE Matérn** Gaussian-random-field surfaces fit over nearest calibration wells with a multi-marker ensemble, composed with an HMM-style **Viterbi** dynamic-programming refinement and Bayesian smoothness-averaging. Substantially reduced local out-of-fold RMSE over single-marker baselines.

Teaching

University of California, Santa Barbara Santa Barbara, CA

Teaching Assistant *Sept. 2017 – March 2025*

- PSTAT 173: Risk Theory · PSTAT 170: Introduction to Financial Mathematics
- PSTAT 160AB: Applied Stochastic Processes · PSTAT 174/274: Time Series Analysis
- PSTAT 126: Regression Analysis · PSTAT 120AB: Probability and Mathematical Statistics
- PSTAT 5A/109: Introduction to Statistics

Graduate Student Mentor *Dec. 2019 – June 2024*

- Supervised undergraduate projects in Financial Mathematics and Optimization.
- Guided students in preparing posters and reports.

Professional Experience

Epic Systems Madison, WI

Technical Services Engineer *Sept. 2014 – Sept. 2015*

- Supported customers in the use of their Electronic Medical Records system.
- Diagnosed customer requests and tailored software to fit their needs.
- Developed and implemented improvements to the Epic codebase.

Prozess Technologie St. Louis, MO

Intern *May 2013 – May 2014*

- Created computational simulations to explore the effectiveness of laboratory equipment.
- Tested, calibrated, and validated laboratory equipment used in pharmaceutical manufacturing.

Fellowships and Awards

Regents Fellowship, *UC Santa Barbara* *2017–2018*

John M. Olin Prize for Excellence in Economics, *Washington University in St. Louis* *2014*

Skills

Quantitative Methods: Quantitative Research, Convex & Portfolio Optimization, Factor Models, Asset Pricing (options & bonds), Black-Litterman, Risk & Volatility Modeling, Time Series Analysis, Signal Processing, Stochastic Analysis, Asymptotic Theory.

Machine Learning: Neural Networks, Gaussian Processes, Reinforcement Learning (Q-Learning), Ensemble Methods (Boosting/Bagging), Bayesian Inference, Time Series Forecasting, Feature Engineering.

Data & Modeling: Statistical Modeling, Experimentation & A/B Testing, Causal Inference, High-Dimensional Regression, Multivariate Statistics, End-to-End Pipelines, Model Deployment.

Languages & Tools: Python (NumPy, SciPy, Pandas, Scikit-Learn, PyTorch, CVXPY), R, MATLAB, SQL, C, Java, JavaScript/NodeJS, FastAPI, Flask, REST APIs, Docker, Git, Linux, AWS, MongoDB, Intersystems Caché, LaTeX, Markdown.